

Two positive components and the intervening dip

Independent observed-data study, version 5.2.0

10 September 2026

What this study finds

The largest two-component improvement in the declared maximal-available-dataset search is at 81 and 92 MeV in 2016 + 2021, with $\Delta_{2/1} = 4.37$. Here $\Delta_{2/1}$ compares the pair with the best single peak searched throughout the same fitted region. Across all seven diagnostic scopes, the largest improvement is 82/91 MeV in 2016 ($\Delta_{2/1} = 19.57$). These are selected locations within the finite catalogue, not global significances.

The largest 2016 feature is strongly dependent on the background model: its existing unfluctuated stress continuum produces a larger two-component endpoint improvement than the data. Combining campaigns also exposes rate disagreement. The historical 66/78 MeV pair receives a smaller improvement, and the new GP-interest regions do not become strong two-component evidence in this common-region fit.

No new toys or random Gaussian fields were generated. This report implements the observed-data portion of Section 5 of the signal-assessment plan [1], with original analysis bins used once per likelihood. It contains extraction displays, a catalogue, fixed-pair reference probability curves, joint amplitude profiles, and deterministic forward predictions of the moving-mask peak–dip shape.

Leading pairs in each diagnostic scope

Scope	m_1/m_2 (MeV)	Q_2	q_{add}	$\Delta_{2/1}$	$p_{2,\text{ref}}$
2015	65/74	13.52	6.86	6.77	5.76×10^{-4}
2016	82/91	32.32	20.00	19.57	4.15×10^{-8}
2021 (10%)	76/81	15.43	11.61	7.71	1.97×10^{-4}
2015 + 2016	81/90	29.22	15.22	15.22	1.98×10^{-7}
2015 + 2021	76/81	13.99	10.30	6.75	4.08×10^{-4}
2016 + 2021	81/92	9.87	4.67	4.37	2.91×10^{-3}
All three	81/90	6.45	4.46	3.52	1.88×10^{-2}

Q_2 tests any positive component against background only; q_{add} compares with the better peak at the two fixed endpoints. $\Delta_{2/1}$ uses the broader single-peak control. The displayed $p_{2,\text{ref}}$ is a fixed-pair asymptotic reference for Q_2 ; it is neither a probability for two particles nor a corrected probability for this catalogue.

Reading the result

A second fitted component can absorb a background interpolation residual, a wider single feature, or a mismatch between campaign rates. The two-peak fits do not resolve the existing 2016 background and state qualifications. Independent background tests and a repeated-experiment or independent-data test are still needed before a discovery interpretation.

1. Common count model and declared search

The inputs are the full 2015 and 2016 spectra and the released native 2021 10% spectrum. The original analysis widths are 0.25 MeV for 2015/2016 and 0.625 MeV for 2021; they are retained throughout. Native finer bins are used only for the existing density-to-yield conversion. No new data increment is opened.

Dataset	Mass hypotheses (MeV)	GP support (MeV)	Analysis bins (MeV)
2015	19–90	14–135	0.25
2016	39–180	30–210	0.25
2021 (10%)	50–250	36–300	0.625

For a fixed pair, campaign d uses all bin centers in

$$\Omega_d = [m_1 - 3\sigma_d(m_1), m_2 + 3\sigma_d(m_2)].$$

The full interval, including the intervening dip, is excluded from GP training. The same bins, training set, GP mean and covariance enter every hypothesis. At least three bins in each external sideband are required. The archived kernel at the integer midpoint rounded down is retained; there is no kernel optimization. This is an exploratory changed-mask policy, not a new qualification of the production background.

The common-region exclusion reduces interpolation information: at the displayed candidate peak bins, the GP prior standard deviations are about 2.3–4.3 times their original moving-mask values for the three individual-scope leaders. The exact comparison includes the declared kernel change and is recorded in `information_loss.csv`; it is not a measured sensitivity or coverage loss.

The fitted expectation is

$$\lambda_{di} = b_{di} + (L_d \theta_d)_i + \psi_1 K_d^{\text{eff}}(m_1) w_{di}(m_1) + \psi_2 K_d^{\text{eff}}(m_2) w_{di}(m_2), \quad \psi_1, \psi_2 \geq 0.$$

The negative log likelihood is the Poisson count term plus $\frac{1}{2} \sum_d \theta_d^T \theta_d$. Signal templates are bin-integrated Gaussians with the recorded resolution; they retain full-support yield normalization. Each coupling-like strength ψ_j is shared across campaigns, while the two strengths are independent of one another. The recorded radiative penalties and observed-density conversions are fixed. Their uncertainties and correlations are not newly profiled here.

Hypothesis	Positive components
H_0	None
H_L	m_1 only
H_H	m_2 only
H_{LH}	Both, fitted simultaneously

The catalogue contains 4,414 scope/pair fits at 1,480 distinct pairs. The initial grid uses even integer masses plus original domain endpoints, with separations 4–24 MeV. Integer refinements surround the leading endpoint-improvement pairs and the prior GP-interest regions. Explicit 90/117 and 92/119 MeV controls and their neighbors extend the catalogue beyond 24 MeV. Every retained pair is evaluated in every eligible scope.

The maximal-available union uses all campaigns that support *both* masses. Thus 2015 drops out when either mass exceeds 90 MeV. The seven scopes overlap in data; choosing their largest improvement is an additional selection. Neither the bounded separation scan nor its local refinements prove a maximum over all possible masses and shapes.

2. What the likelihood comparisons and p-values mean

Write $\ell_0, \ell_L, \ell_H, \ell_{LH}$ for the four minimized negative log likelihoods. We record

$$Q_2 = 2(\ell_0 - \ell_{LH}), \quad q_{\text{add}} = 2[\min(\ell_L, \ell_H) - \ell_{LH}].$$

A large Q_2 can be caused by one component. The endpoint statistic q_{add} requires both fixed templates to improve the fit, but can favor two templates that approximate a single peak between them.

A broader single-peak control

For every pair, we therefore scan a single nonnegative component across the union of the campaigns' fitted mass intervals, intersected with the scope's original search support. All likelihood bins and the background constraint remain fixed. Every integer mass and the exact interval endpoints are tested; the best mass is refined within a one-MeV neighborhood to 0.001 MeV numerical tolerance using the resolution function and native-bin density. This is a local refinement of the best integer solution, not a proof of a continuous global optimum.

$$\Delta_{2/1}^{\text{signed}} = 2(\ell_{1,\text{best}} - \ell_{LH}).$$

The families are not nested, so the signed value can be negative. Catalogue ranking uses its positive part; signed values and the earlier intervening-only control remain in the CSV. No calibrated p-value is attached to $\Delta_{2/1}$. This control was added during the exploratory review and is recorded as such in the protocol.

Fixed-pair asymptotic reference for Q_2

The efficient null information is

$$I = S^T W S - S^T W L (1 + L^T W L)^{-1} L^T W S, \quad W = \text{diag}(1/\hat{\lambda}_0).$$

With $c = I_{12}/\sqrt{I_{11}I_{22}}$, the two-boundary quadratic reference [4] has

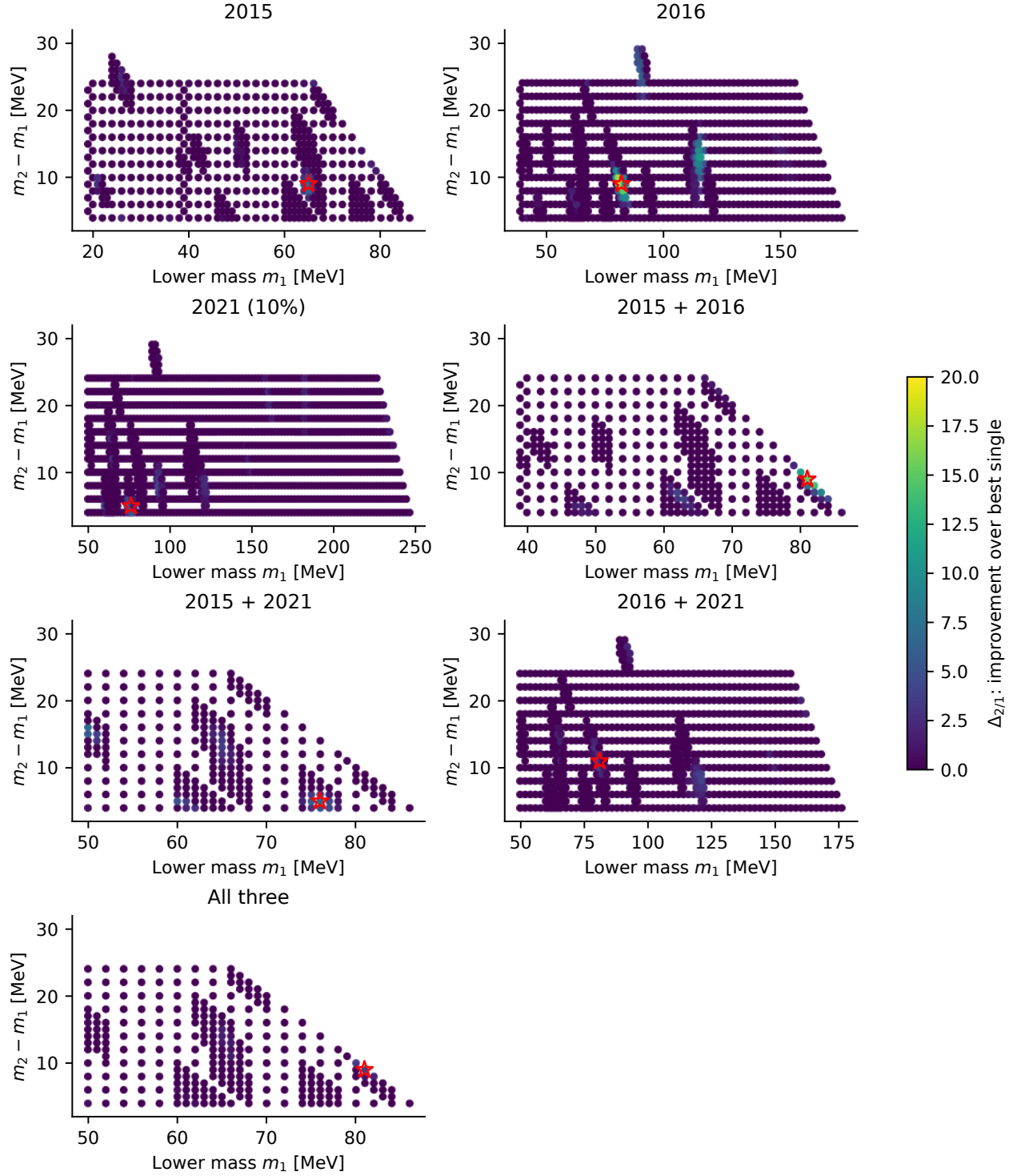
$$w_2 = \frac{\arccos(c)}{2\pi}, \quad w_1 = \frac{1}{2}, \quad w_0 = \frac{1}{2} - w_2, \quad Q_2 \sim w_0 \delta_0 + w_1 \chi_1^2 + w_2 \chi_2^2.$$

For $Q_2 > 0$, $p_{2,\text{ref}} = \frac{1}{2} \Pr(\chi_1^2 \geq Q_2) + w_2 \Pr(\chi_2^2 \geq Q_2)$; at zero we use the inclusive-tail convention $p = 1$. These are fixed-pair asymptotic references under the specified likelihood. The auxiliary-GP interpretation and marginal tails are not newly calibrated. The background-profiled null information, rather than the archived moving-mask score correlation, determines these weights.

No global or independent-dip probability

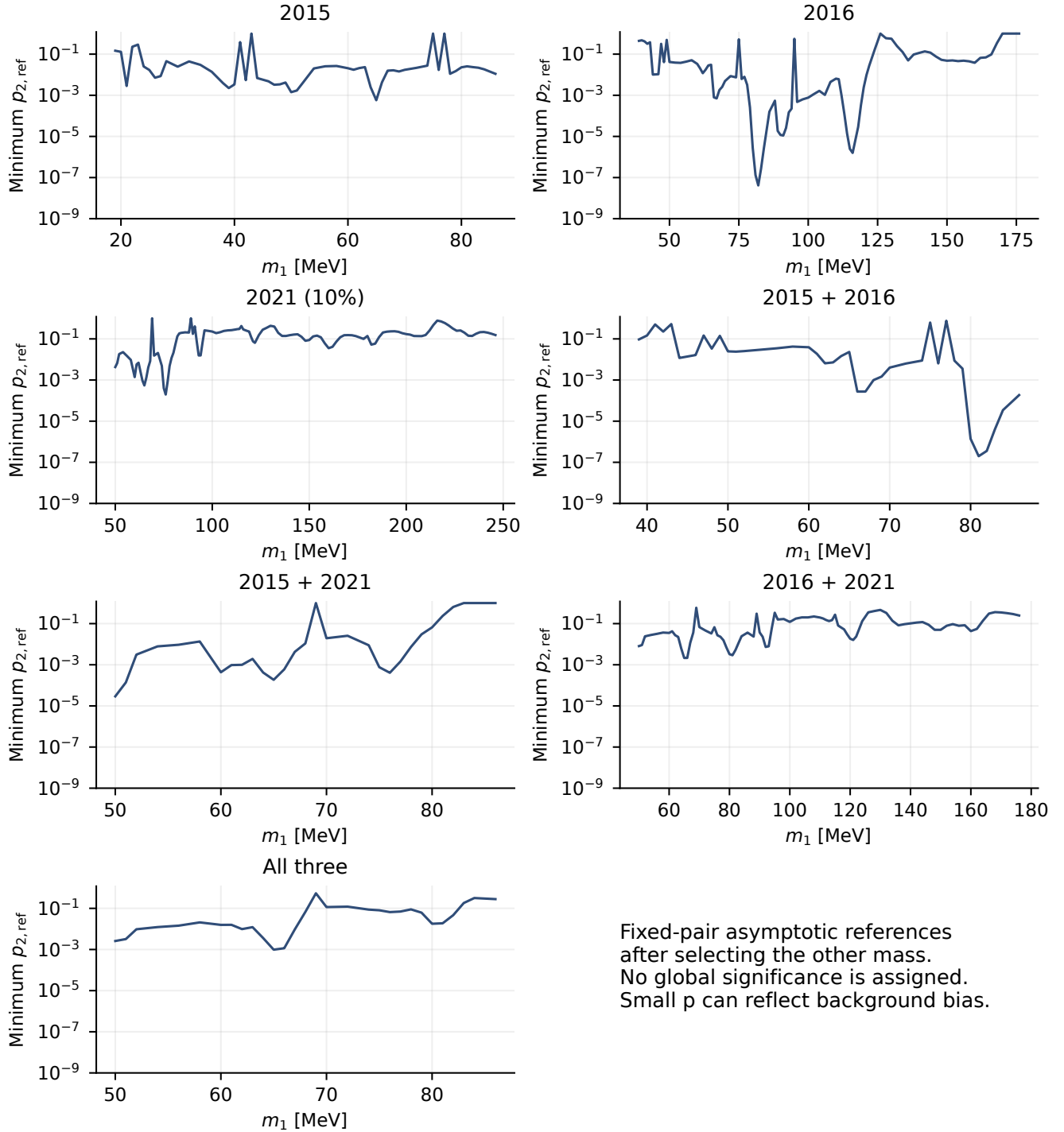
Mass selection, the second mass, scope choice, and the analysis development enlarge the search [5,6]. The old single-mass GP probability or trials factor cannot correct the new pair search. The saved $\exp(-q_{\text{add}}/2)$ column is only a conservative bound in a fixed two-coordinate Gaussian experiment; it is not used to assign significance to the broader single-peak comparison. The dip never enters as a second independent likelihood factor.

3. Survey of two-component improvements



Each point is an evaluated pair. Color gives the positive improvement over the best single peak in the full fixed region; red stars mark each scope leader. The same color scale is used throughout. Refined and wider targeted pairs are included. These are selected observed improvements, not global significance contours.

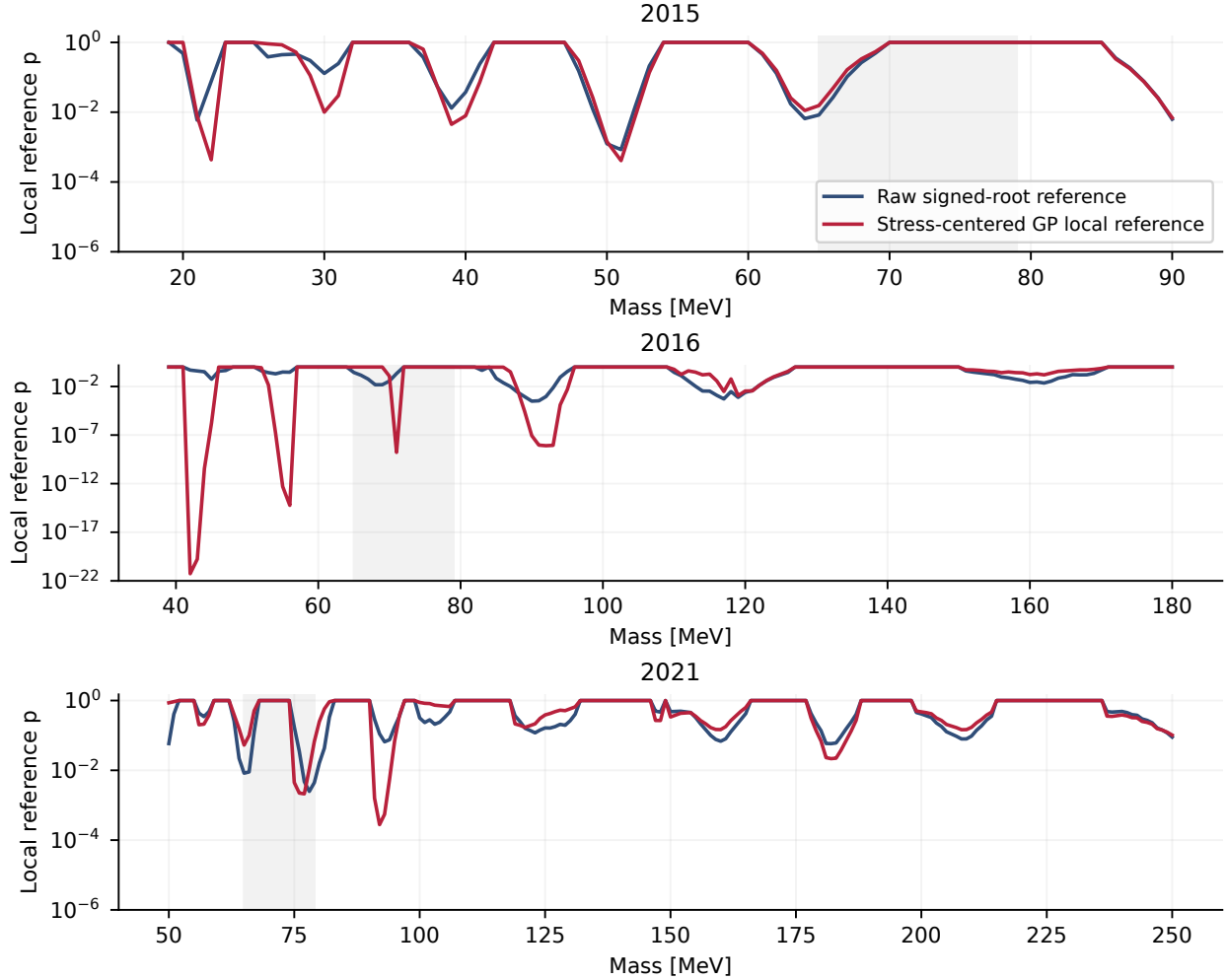
4. Reference p-value curves for the pair model



At each lower mass, the plotted value is the smallest fixed-pair Q_2 reference p-value among tested upper masses. This selection is shown explicitly and has no trials correction. The probability concerns any positive component, whereas the catalogue ranks the comparison with the best single peak. In particular, the small 2016 values do not survive interpretation as evidence under an independently qualified background.

5. Where the new significance-GP model changes the investigation

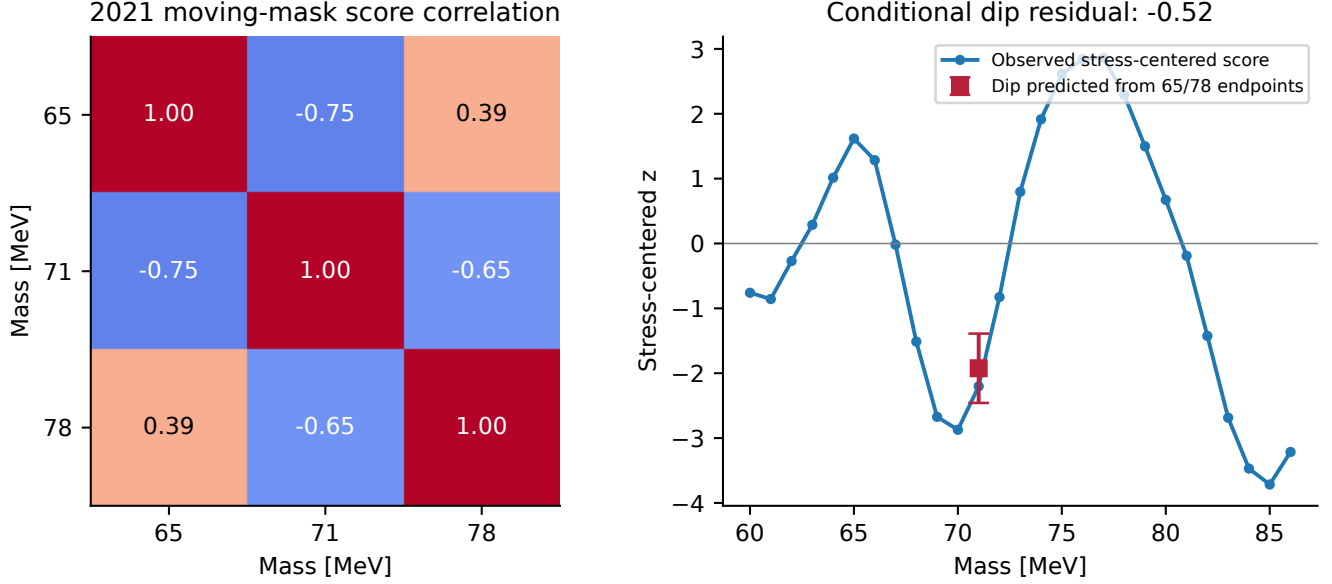
The v5.1.0 and audited v5.0.2 fields distinguish the raw signed root r_m from the stress-centered score $z_m = (r_m - a_m)/s_m$. The conditional local probability also applies the positive-fit gate $r_m > 0$; a negative observed root is not a positive-peak candidate. The leading gated GP locations are 51 MeV in 2015, 42 MeV in 2016, 92 MeV in 2021, and 76 MeV in the combined union. The 2016 and union conditional scores are largely driven by negative stress-reference offsets [2,3].



Previously saved local probability fields, recomputed analytically from their pinned roots and stress standardization. The gray interval marks the historical low/high neighborhood. These curves guide the additional catalogue regions; their probabilities are not transferred to the two-peak statistic.

The new common-region fits explicitly test the 42/55 MeV and 78/92 MeV neighborhoods, along with the historical 65–66/78–79 MeV candidates. A high stress-centered score does not guarantee that two positive count components are needed. For example, the 2016 42/55 MeV endpoint improvement is zero; in 2021 the 78/92 MeV endpoint improvement is only 0.15. All corresponding extraction and forward-response files accompany the report.

6. The intervening dip is strongly correlated with the peaks



The old 2021 moving-mask field has strong negative correlations between the dip and both endpoints. Conditioning on the observed 65/78 MeV endpoint scores predicts most of the 71 MeV deficit. This is a post-selection conditional diagnostic under the archived stress reference, not a particle significance.

Dataset	$R(65, 78)$	$R(65, 71)$	$R(71, 78)$
2015	-0.187	-0.552	-0.608
2016	0.003	-0.643	-0.682
2021	0.387	-0.746	-0.655

For a fixed triple of stress-standardized scores, a descriptive conditional residual is

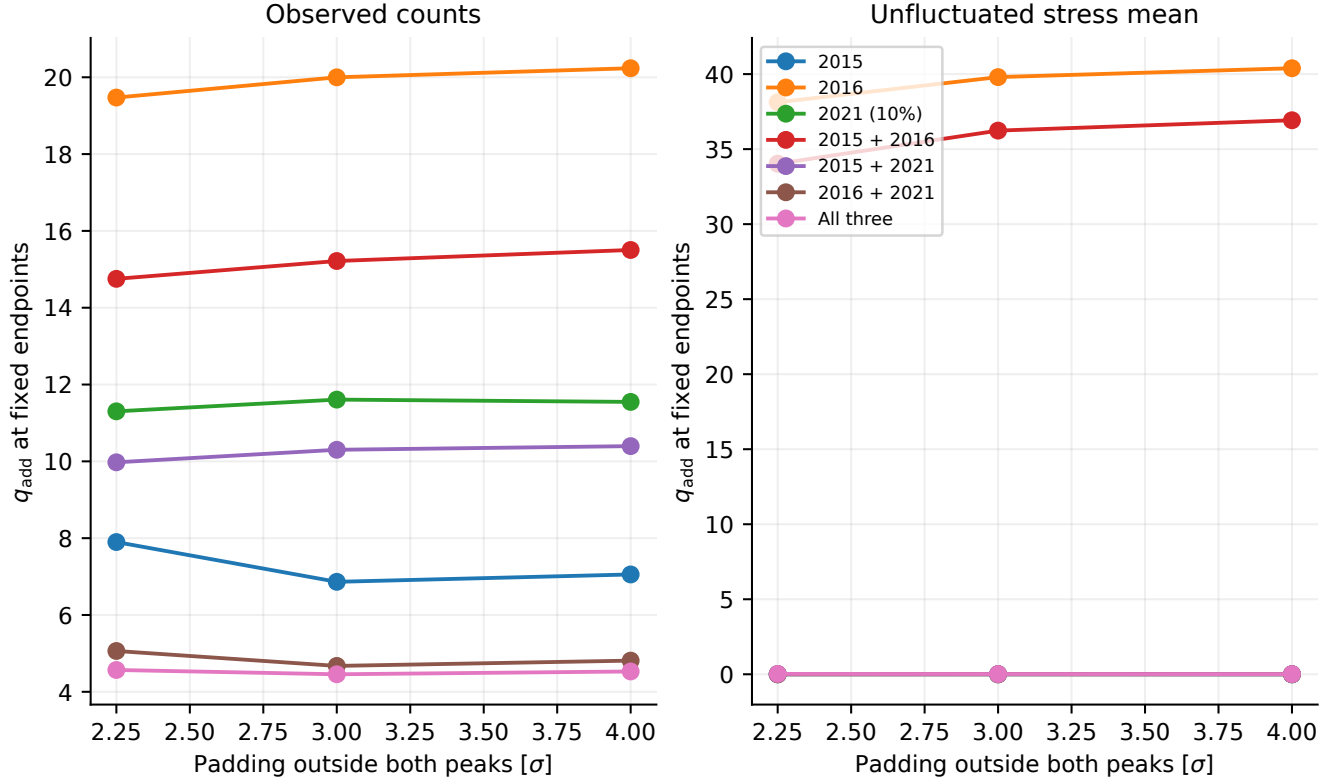
$$u_D = \frac{z_D - R_{DP} R_{PP}^{-1} z_P}{\sqrt{R_{DD} - R_{DP} R_{PP}^{-1} R_{PD}}}.$$

For 2021 at 65/71/78 MeV, the endpoints predict $z_{71} = -1.924$ with conditional width 0.535; the observed value gives $u_D = -0.520$. Multiplying three local probabilities would therefore count correlated structure as independent evidence.

The old significance-field covariance is distinct from both the background covariance in the new joint fit and the new fitted-amplitude covariance. In the new two-component fit, the dip can constrain the common background and thereby correlate the two amplitudes. The covariance of those amplitudes is retained in the profile surfaces later in this report.

Independent Poisson input bins can produce strong correlations across mass hypotheses because each moving mask reuses events and sidebands [7]. Detector mass smearing alone does not establish correlated count noise. The aggregate histograms cannot test multiple candidates per event, shared weights, or acquisition-block dependence. The illustrative count-correlation test on the next page is kept separate from the measured input arrays.

7. Background, window, and rate checks



All three padding choices are shown for every scope leader, with the candidate masses fixed. The right panel fits the existing smooth stress means without fluctuations. These are deterministic estimator responses, not toy distributions or p-values. The endpoint statistic is used so that all four nested hypotheses remain directly comparable.

The 2016 82/91 MeV data fit gives $q_{\text{add}} \simeq 20.0$, while its archived no-signal stress mean gives about 39.8. Widening the mask does not remove the discrepancy. The 2016 stress construction has its own source-convergence and qualification limitations; this calculation identifies strong model dependence, rather than supplying an accepted physical null or an automatic background subtraction.

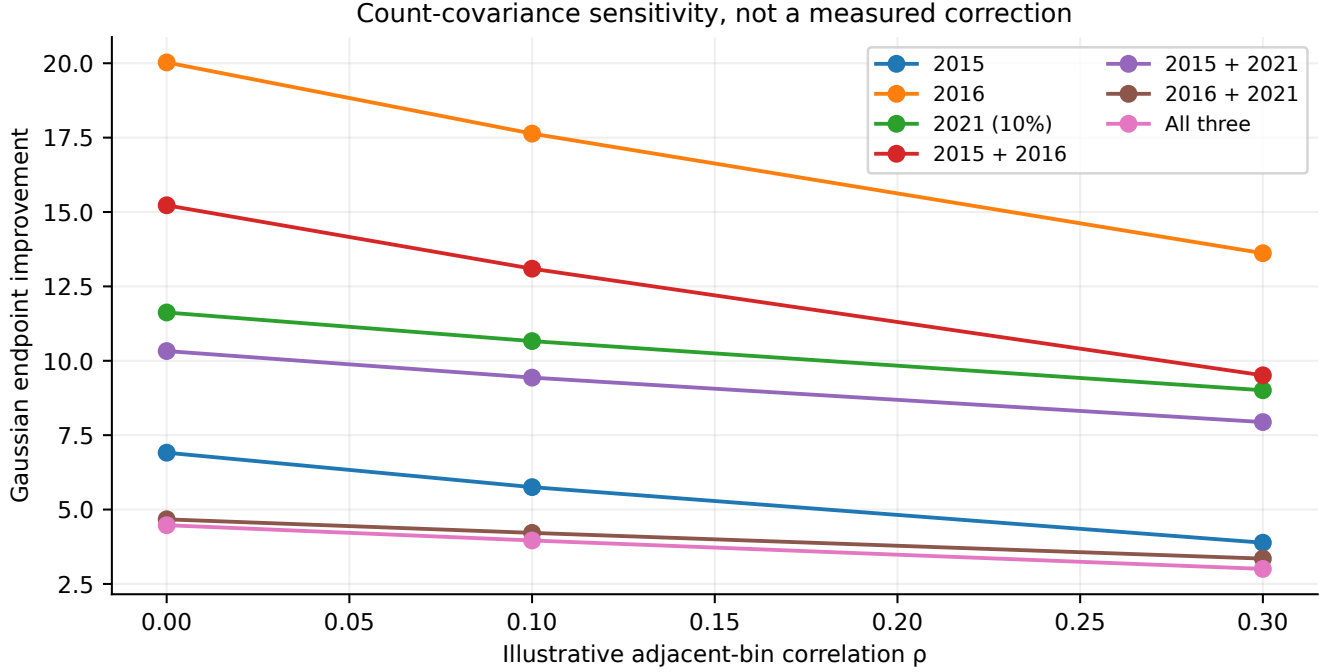
The union leader is also sensitive to the common-rate hypothesis. The descriptive rate statistic compares the shared two-strength fit with two strengths fitted independently in each campaign:

$$D_{\text{rate}} = 2 \left[\ell_{LH}^{\text{shared}} - \sum_d \ell_{LH,d}^{\text{independent}} \right].$$

No p-value is assigned: masses were selected, amplitudes can be on boundaries, and the normalization/systematic model is fixed. Large values nevertheless show that summing positive structures across campaigns is not equivalent to a common-coupling particle model.

Scope leader	D_{rate}	Amplitude correlation
2015	0.00	0.785
2016	0.00	0.662
2021 (10%)	0.00	0.567
2015 + 2016	4.88	0.670
2015 + 2021	1.44	0.563
2016 + 2021	16.16	0.235
All three	29.66	0.499

8. Illustrative correlation between input count bins



A Gaussian residual sensitivity calculation with fixed common-region mean and GP covariance, using $V = C + \text{diag}(\sqrt{b})Q \text{diag}(\sqrt{b})$ and $Q_{ij} = \rho^{|i-j|}$. These values are not replacement Poisson fits or corrected HPS significances.

This calculation retains the change in amplitude variance. Positive neighboring-bin correlation generally reduces the amount of apparently independent shape information in these examples. The supplied values $\rho = 0, 0.1, 0.3$ are illustrative and are not estimated from HPS events. Equal neighboring-bin correlation also refers to different physical bin widths across campaigns.

The existing GP prediction uncertainty is already included through C . It is not added again as an extra measured count covariance. An event-level count-covariance model would require candidate/event identifiers or a generator preserving the actual event structure. No such information is inferred from the stored histogram variances.

How to use the following catalogue

Each of the seven pages lists five separated high-improvement candidates, rather than five nearly identical neighboring mass points. Candidates are retained greedily in descending $\Delta_{2/1}$ with a Manhattan separation of at least 6 MeV in the two masses. This is only a readable catalogue rule; all 4,414 evaluated rows are available digitally.

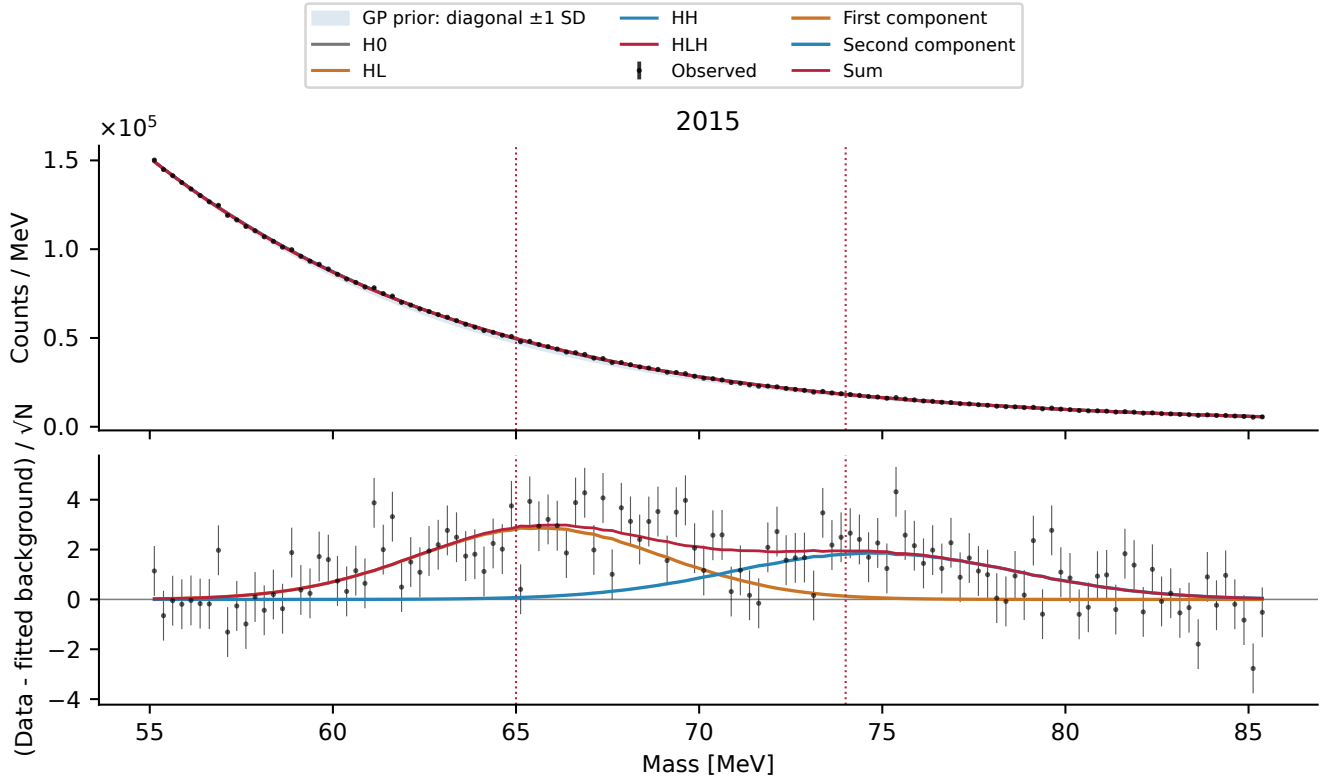
The extraction display is centered on that scope's leading pair. Its blue band is the diagonal GP prior uncertainty from the shared external training set, not a post-fit confidence band. The lower panel shows $(N - \hat{b}_{LH})/\sqrt{N}$ with the two fitted signal components and their sum. Its error bars show Poisson count errors only; the fitted background is uncertain and correlated across bins, so this display is not an independent significance measurement.

The listed ψ values use the retained full-yield conversion, and the two-peak correlation uses the unconstrained observed curvature at the pair fit. It need not equal minus the null information correlation used for the chi-bar reference. A constrained zero amplitude and a curvature ellipse that crosses the boundary do not define an ordinary confidence interval.

9. Candidate catalogue: 2015

The leading pair is 65/74 MeV. The best single mass in the same region is 64.716 MeV, giving $\Delta_{2/1} = 6.77$. The fitted strengths are $\psi_1 = 60.333 \times 10^{-6}$ and $\psi_2 = 63.596 \times 10^{-6}$.

Rank	m_1/m_2 (MeV)	Q_2	q_{add}	$\Delta_{2/1}$	$p_{2,\text{ref}}$
1	65/74	13.52	6.86	6.77	5.76×10^{-4}
2	26/50	7.71	2.91	2.80	8.53×10^{-3}
3	21/30	9.96	2.84	2.72	2.84×10^{-3}
4	66/90	7.73	2.23	1.80	7.16×10^{-3}
5	78/90	7.64	1.71	1.71	1.11×10^{-2}



Original analysis-bin counts and the four common-region hypotheses. The two vertical markers give the selected masses. Every hypothesis uses the same count bins and external GP constraint. Separate campaign panels are likelihood constituents, not repeated measurements of the same events.

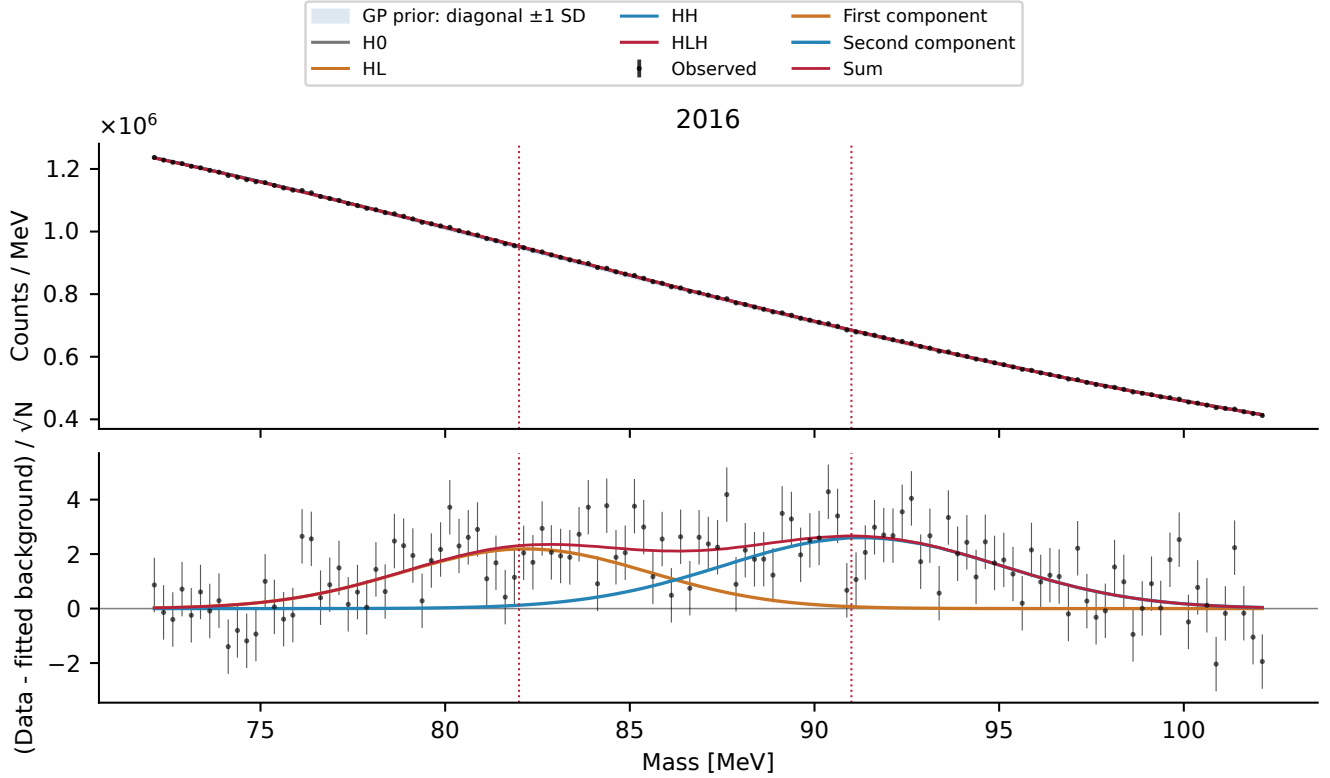
Campaign	Fit region (MeV)	N_1 shared	N_2 shared	N_1 independent	N_2 independent
2015	54.9–85.5	10593	4750	10593	4750

For this pair, the fitted-amplitude correlation is 0.785 and the rate-comparison deviance is 0.00. The fit region and masses are selected diagnostic choices; the table is not a ranking of particle probabilities.

10. Candidate catalogue: 2016

The leading pair is 82/91 MeV. The best single mass in the same region is 90.394 MeV, giving $\Delta_{2/1} = 19.57$. The fitted strengths are $\psi_1 = 15.290 \times 10^{-6}$ and $\psi_2 = 21.421 \times 10^{-6}$.

Rank	m_1/m_2 (MeV)	Q_2	q_{add}	$\Delta_{2/1}$	$p_{2,\text{ref}}$
1	82/91	32.32	20.00	19.57	4.15×10^{-8}
2	116/129	25.08	11.21	10.81	1.58×10^{-6}
3	91/116	20.36	6.50	6.17	1.11×10^{-5}
4	114/124	17.70	8.39	4.24	5.90×10^{-5}
5	62/69	7.53	3.00	2.79	1.19×10^{-2}



Original analysis-bin counts and the four common-region hypotheses. The two vertical markers give the selected masses. Every hypothesis uses the same count bins and external GP constraint. Separate campaign panels are likelihood constituents, not repeated measurements of the same events.

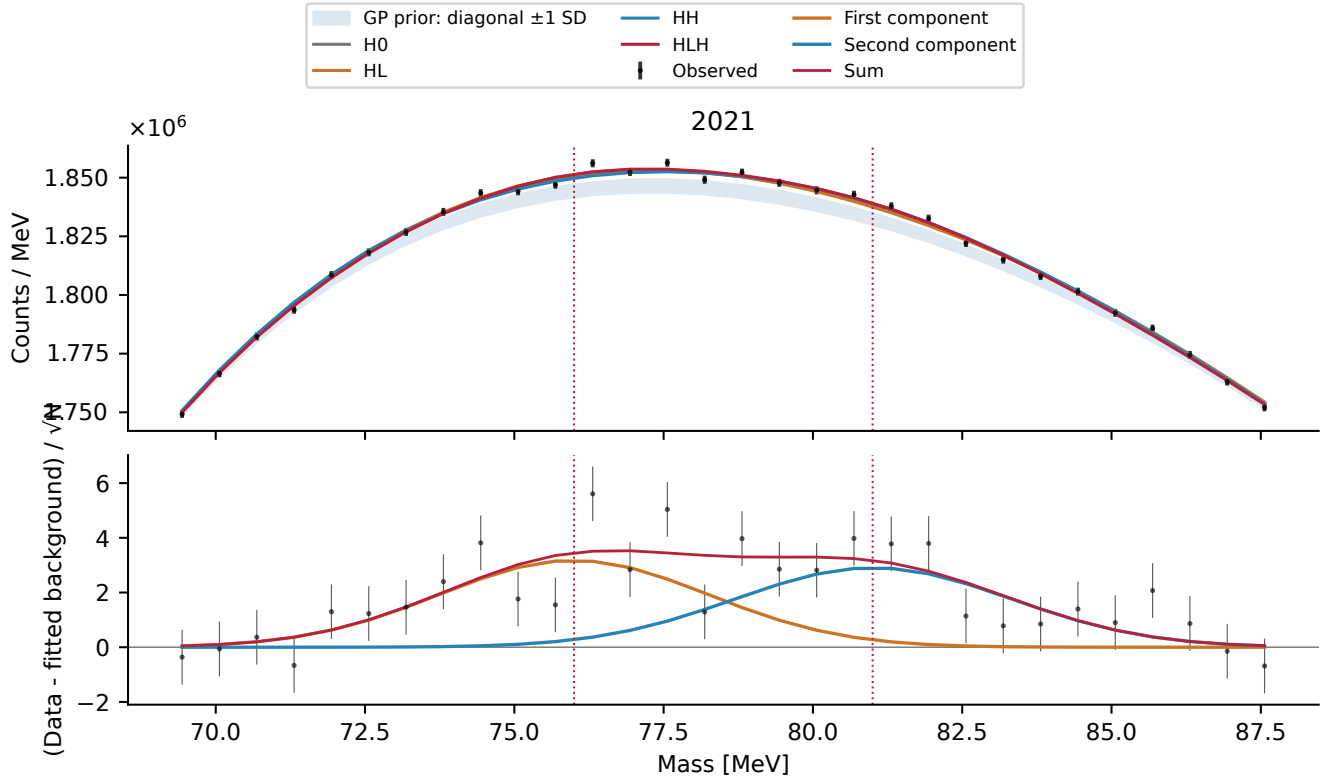
Campaign	Fit region (MeV)	N_1 shared	N_2 shared	N_1 independent	N_2 independent
2016	72.0–102.2	35832	40273	35832	40273

For this pair, the fitted-amplitude correlation is 0.662 and the rate-comparison deviance is 0.00. The fit region and masses are selected diagnostic choices; the table is not a ranking of particle probabilities.

11. Candidate catalogue: 2021 (10%)

The leading pair is 76/81 MeV. The best single mass in the same region is 78.344 MeV, giving $\Delta_{2/1} = 7.71$. The fitted strengths are $\psi_1 = 7.134 \times 10^{-6}$ and $\psi_2 = 6.309 \times 10^{-6}$.

Rank	m_1/m_2 (MeV)	Q_2	q_{add}	$\Delta_{2/1}$	$p_{2,\text{ref}}$
1	76/81	15.43	11.61	7.71	1.97×10^{-4}
2	60/65	11.77	5.54	5.34	1.39×10^{-3}
3	94/102	6.92	4.91	4.53	1.57×10^{-2}
4	65/78	12.60	3.84	3.51	5.44×10^{-4}
5	122/130	4.16	2.82	2.72	6.42×10^{-2}



Original analysis-bin counts and the four common-region hypotheses. The two vertical markers give the selected masses. Every hypothesis uses the same count bins and external GP constraint. Separate campaign panels are likelihood constituents, not repeated measurements of the same events.

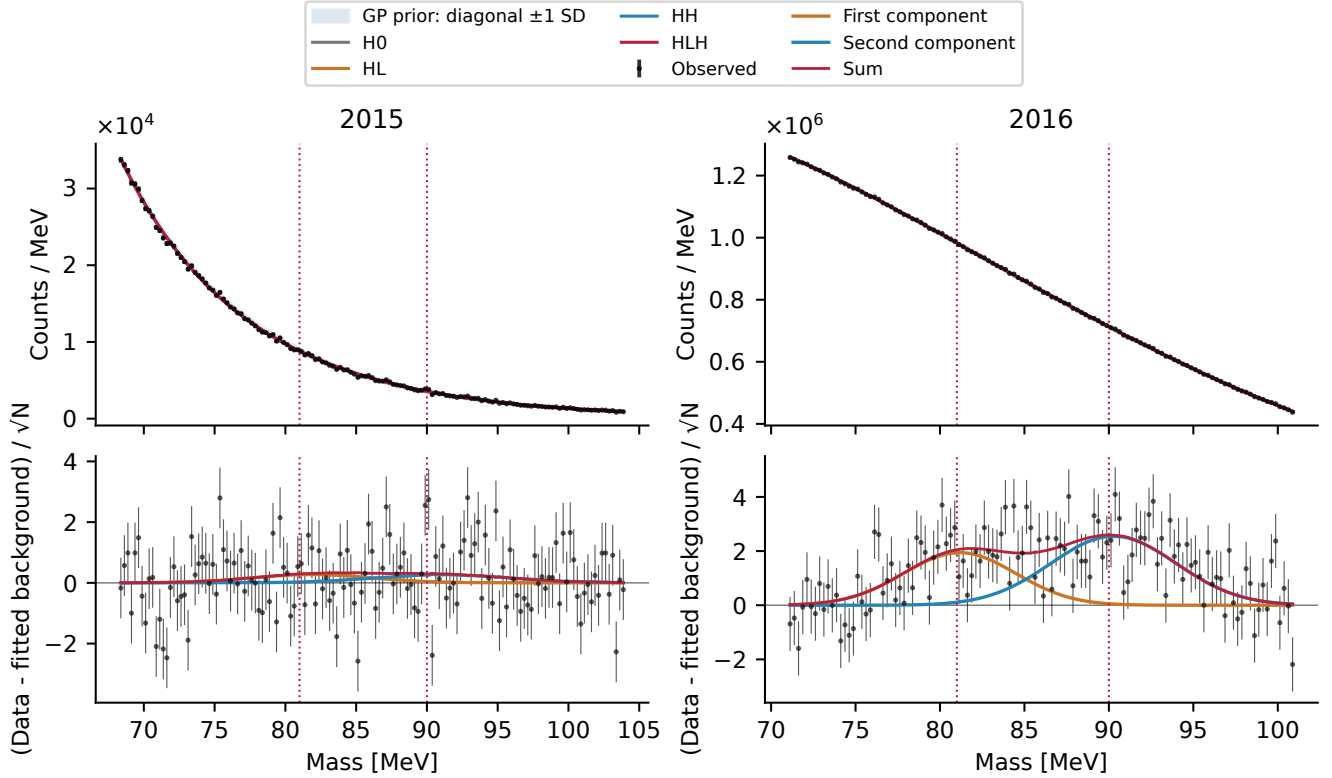
Campaign	Fit region (MeV)	N_1 shared	N_2 shared	N_1 independent	N_2 independent
2021	69.3–87.9	30793	28865	30793	28865

For this pair, the fitted-amplitude correlation is 0.567 and the rate-comparison deviance is 0.00. The fit region and masses are selected diagnostic choices; the table is not a ranking of particle probabilities.

12. Candidate catalogue: 2015 + 2016

The leading pair is 81/90 MeV. The best single mass in the same region is 90.000 MeV, giving $\Delta_{2/1} = 15.22$. The fitted strengths are $\psi_1 = 13.343 \times 10^{-6}$ and $\psi_2 = 20.552 \times 10^{-6}$.

Rank	m_1/m_2 (MeV)	Q_2	q_{add}	$\Delta_{2/1}$	$p_{2,\text{ref}}$
1	81/90	29.22	15.22	15.22	1.98×10^{-7}
2	62/69	8.70	4.56	3.97	6.49×10^{-3}
3	48/53	3.24	2.48	1.96	1.12×10^{-1}
4	68/90	11.35	1.55	1.55	1.07×10^{-3}
5	50/68	5.49	0.54	0.34	2.48×10^{-2}



Original analysis-bin counts and the four common-region hypotheses. The two vertical markers give the selected masses. Every hypothesis uses the same count bins and external GP constraint. Separate campaign panels are likelihood constituents, not repeated measurements of the same events.

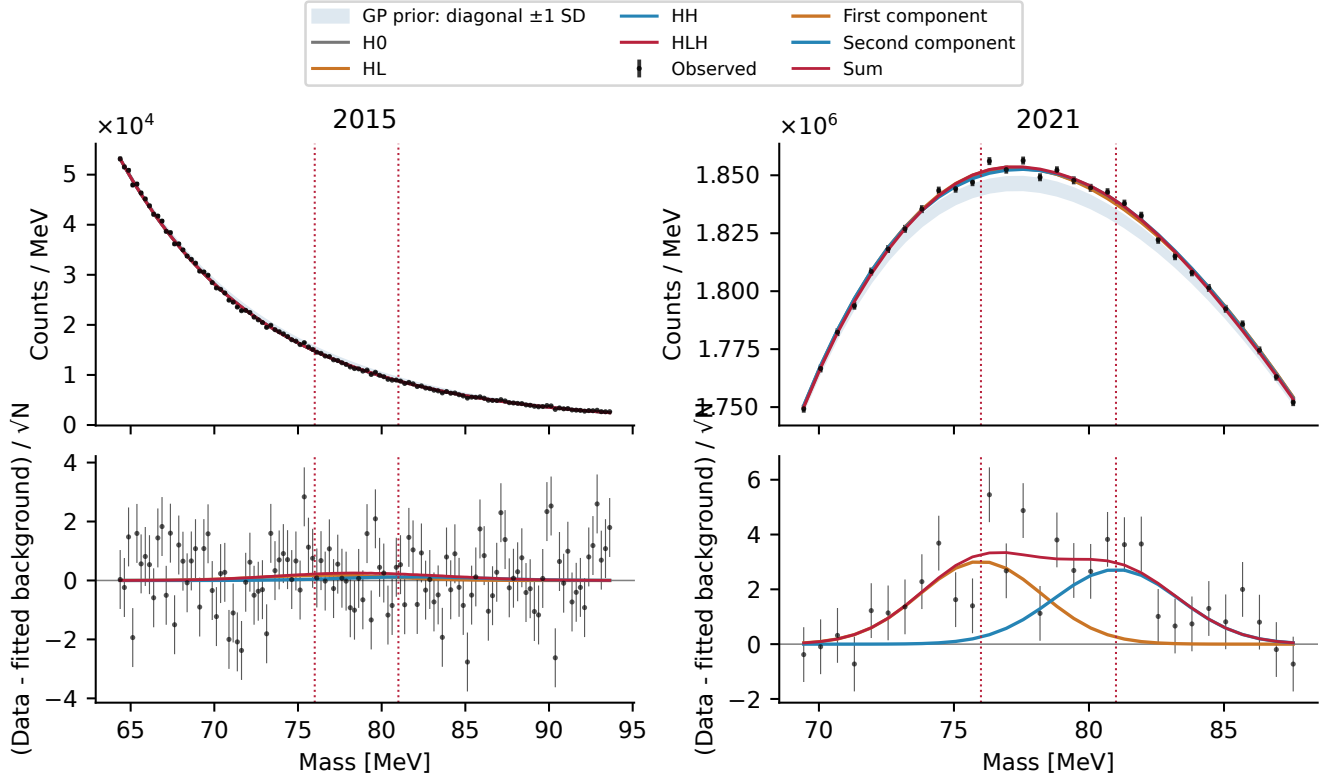
Campaign	Fit region (MeV)	N_1 shared	N_2 shared	N_1 independent	N_2 independent
2015	68.3–104.1	530	370	126	1315
2016	71.1–101.1	31867	39770	32658	39060

For this pair, the fitted-amplitude correlation is 0.670 and the rate-comparison deviance is 4.88. The fit region and masses are selected diagnostic choices; the table is not a ranking of particle probabilities.

13. Candidate catalogue: 2015 + 2021

The leading pair is 76/81 MeV. The best single mass in the same region is 78.244 MeV, giving $\Delta_{2/1} = 6.75$. The fitted strengths are $\psi_1 = 6.777 \times 10^{-6}$ and $\psi_2 = 5.907 \times 10^{-6}$.

Rank	m_1/m_2 (MeV)	Q_2	q_{add}	$\Delta_{2/1}$	$p_{2,\text{ref}}$
1	76/81	13.99	10.30	6.75	4.08×10^{-4}
2	50/66	18.81	7.81	6.55	2.91×10^{-5}
3	60/65	14.04	5.14	5.04	4.32×10^{-4}
4	65/78	14.72	3.31	3.31	1.86×10^{-4}
5	50/54	8.61	0.66	0.66	6.36×10^{-3}



Original analysis-bin counts and the four common-region hypotheses. The two vertical markers give the selected masses. Every hypothesis uses the same count bins and external GP constraint. Separate campaign panels are likelihood constituents, not repeated measurements of the same events.

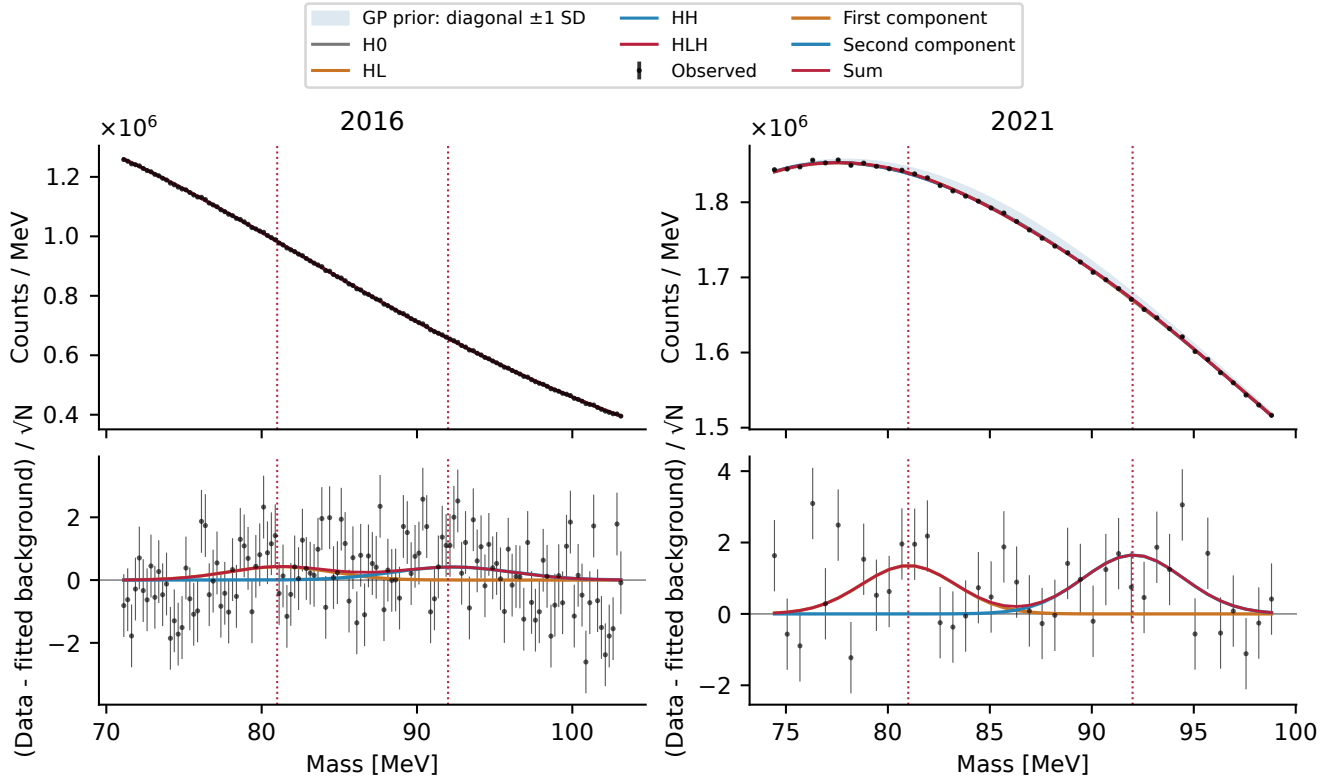
Campaign	Fit region (MeV)	N_1 shared	N_2 shared	N_1 independent	N_2 independent
2015	64.1–93.7	421	235	0	0
2021	69.3–87.9	29255	27026	30793	28865

For this pair, the fitted-amplitude correlation is 0.563 and the rate-comparison deviance is 1.44. The fit region and masses are selected diagnostic choices; the table is not a ranking of particle probabilities.

14. Candidate catalogue: 2016 + 2021

The leading pair is 81/92 MeV. The best single mass in the same region is 92.485 MeV, giving $\Delta_{2/1} = 4.37$. The fitted strengths are $\psi_1 = 2.940 \times 10^{-6}$ and $\psi_2 = 3.515 \times 10^{-6}$.

Rank	m_1/m_2 (MeV)	Q_2	q_{add}	$\Delta_{2/1}$	$p_{2,\text{ref}}$
1	81/92	9.87	4.67	4.37	2.91×10^{-3}
2	121/129	6.80	4.14	3.99	1.60×10^{-2}
3	66/79	10.11	2.90	2.21	2.15×10^{-3}
4	92/120	7.91	2.36	2.00	7.33×10^{-3}
5	160/180	4.64	1.82	1.81	4.32×10^{-2}



Original analysis-bin counts and the four common-region hypotheses. The two vertical markers give the selected masses. Every hypothesis uses the same count bins and external GP constraint. Separate campaign panels are likelihood constituents, not repeated measurements of the same events.

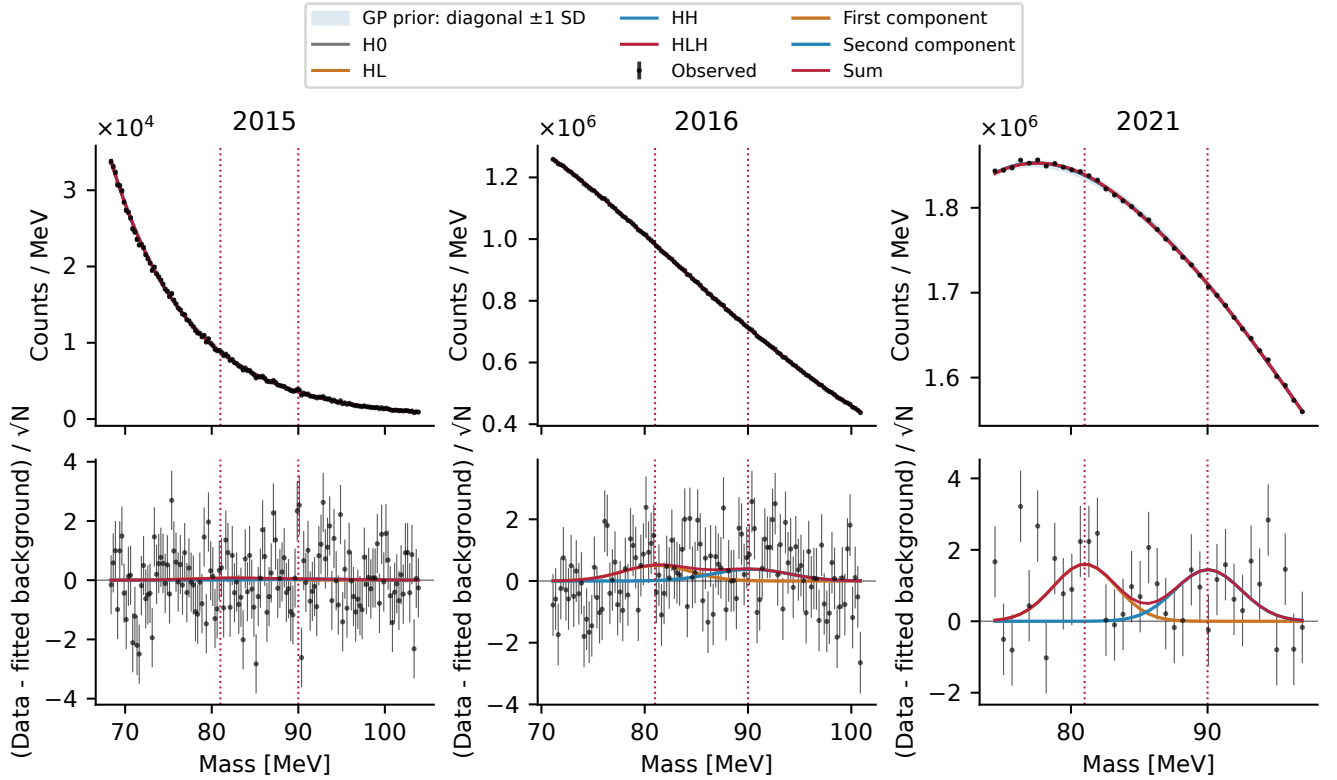
Campaign	Fit region (MeV)	N_1 shared	N_2 shared	N_1 independent	N_2 independent
2016	71.1–103.4	7022	6415	28813	33117
2021	74.1–99.3	13451	16614	11114	8027

For this pair, the fitted-amplitude correlation is 0.235 and the rate-comparison deviance is 16.16. The fit region and masses are selected diagnostic choices; the table is not a ranking of particle probabilities.

15. Candidate catalogue: All three

The leading pair is 81/90 MeV. The best single mass in the same region is 79.789 MeV, giving $\Delta_{2/1} = 3.52$. The fitted strengths are $\psi_1 = 3.473 \times 10^{-6}$ and $\psi_2 = 3.073 \times 10^{-6}$.

Rank	m_1/m_2 (MeV)	Q_2	q_{add}	$\Delta_{2/1}$	$p_{2,\text{ref}}$
1	81/90	6.45	4.46	3.52	1.88×10^{-2}
2	65/80	11.54	2.00	1.74	9.93×10^{-4}
3	50/54	2.01	0.79	0.78	2.04×10^{-1}
4	60/66	6.96	0.70	0.69	1.58×10^{-2}
5	50/66	9.82	0.33	0.33	2.59×10^{-3}

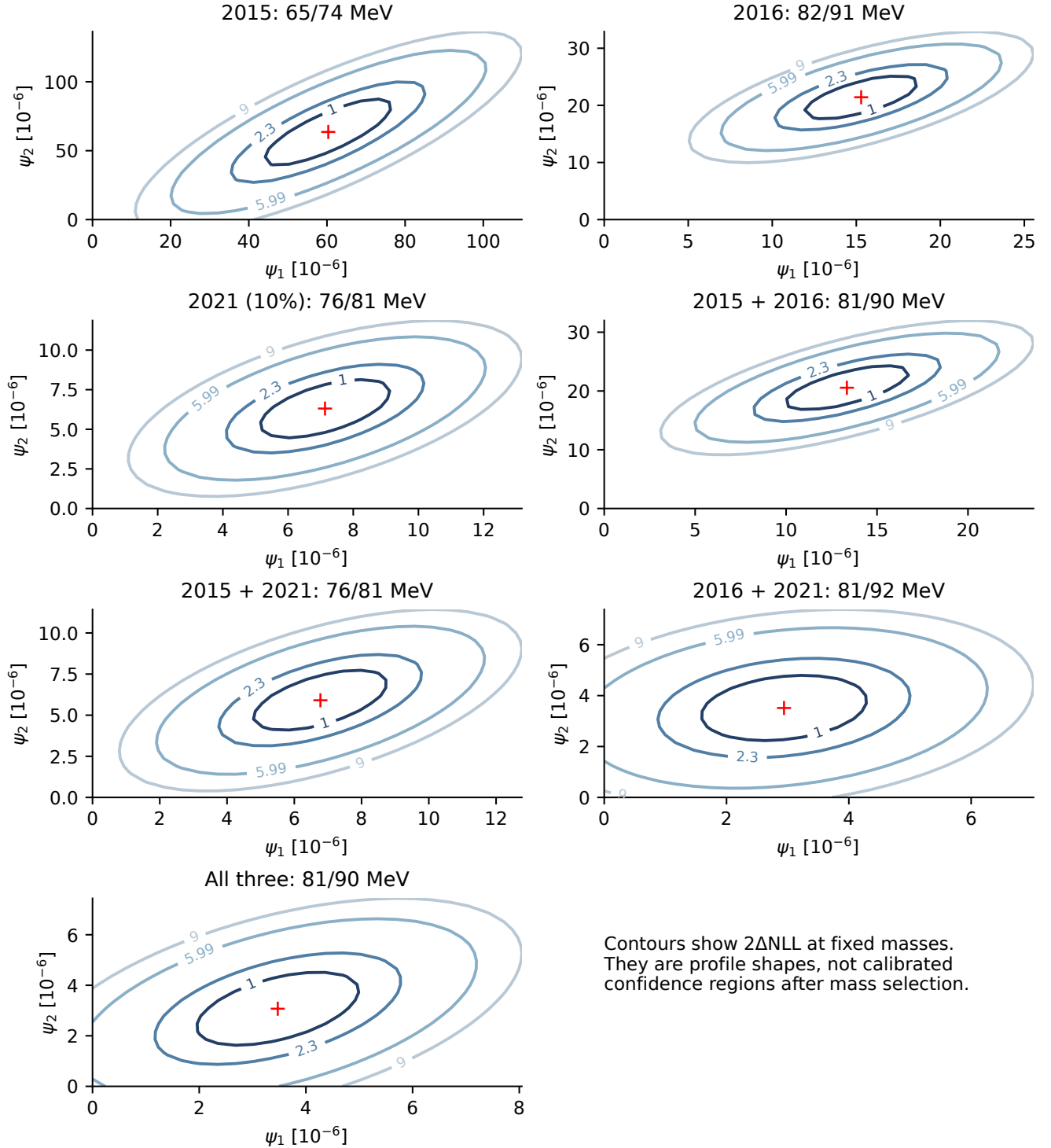


Original analysis-bin counts and the four common-region hypotheses. The two vertical markers give the selected masses. Every hypothesis uses the same count bins and external GP constraint. Separate campaign panels are likelihood constituents, not repeated measurements of the same events.

Campaign	Fit region (MeV)	N_1 shared	N_2 shared	N_1 independent	N_2 independent
2015	68.3–104.1	138	55	126	1315
2016	71.1–101.1	8294	5946	32658	39060
2021	74.1–97.3	15887	14540	10155	249

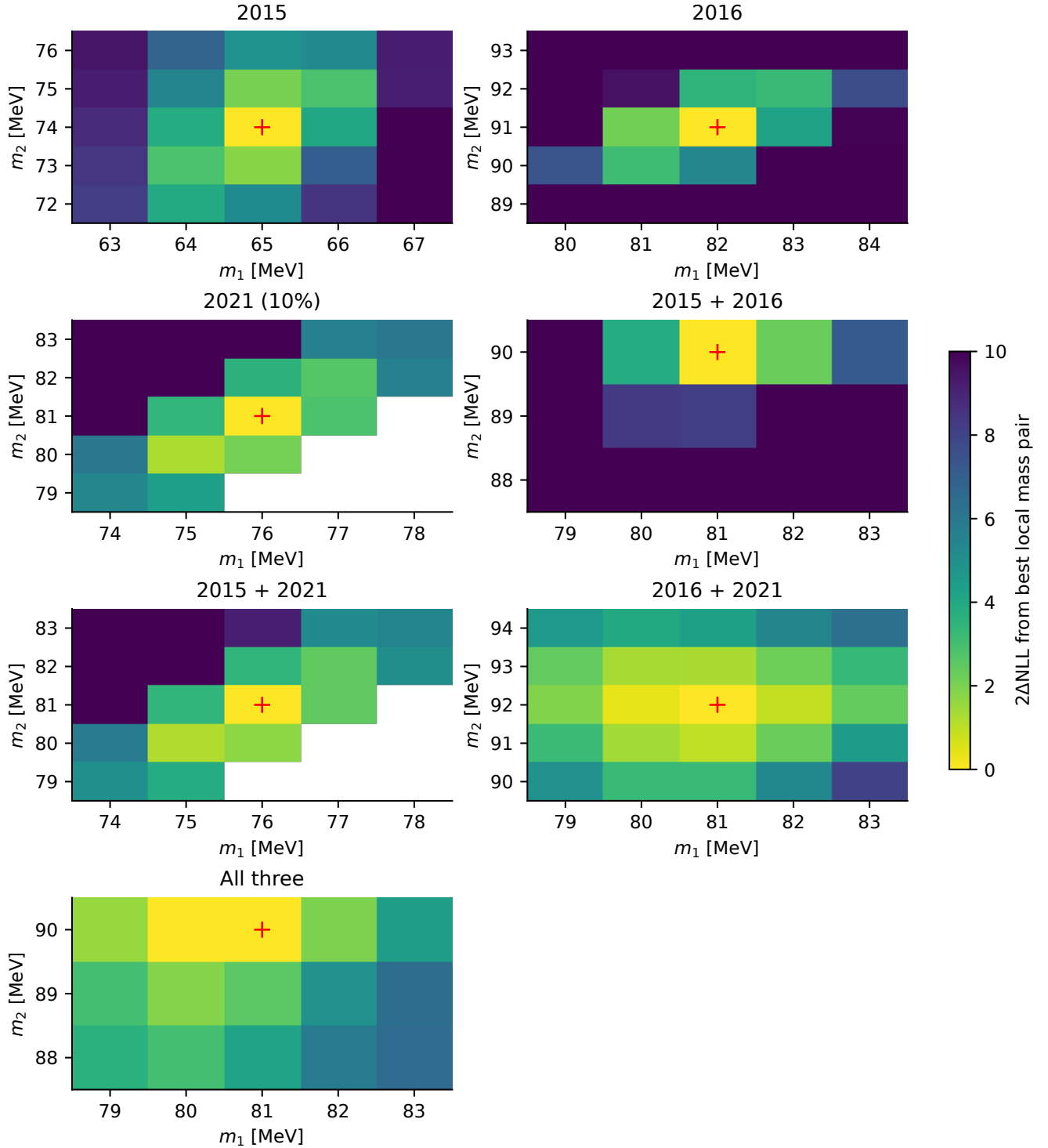
For this pair, the fitted-amplitude correlation is 0.499 and the rate-comparison deviance is 29.66. The fit region and masses are selected diagnostic choices; the table is not a ranking of particle probabilities.

16. Correlated amplitude profile surfaces



Profiles over the Gaussian background nuisances at the seven selected pairs. The contours are levels of twice the negative-log-likelihood change, and the red crosses are the nonnegative joint fits. Curvature covariance matrices and separate component profile traces are saved. No post-selection coverage is assigned to these contours.

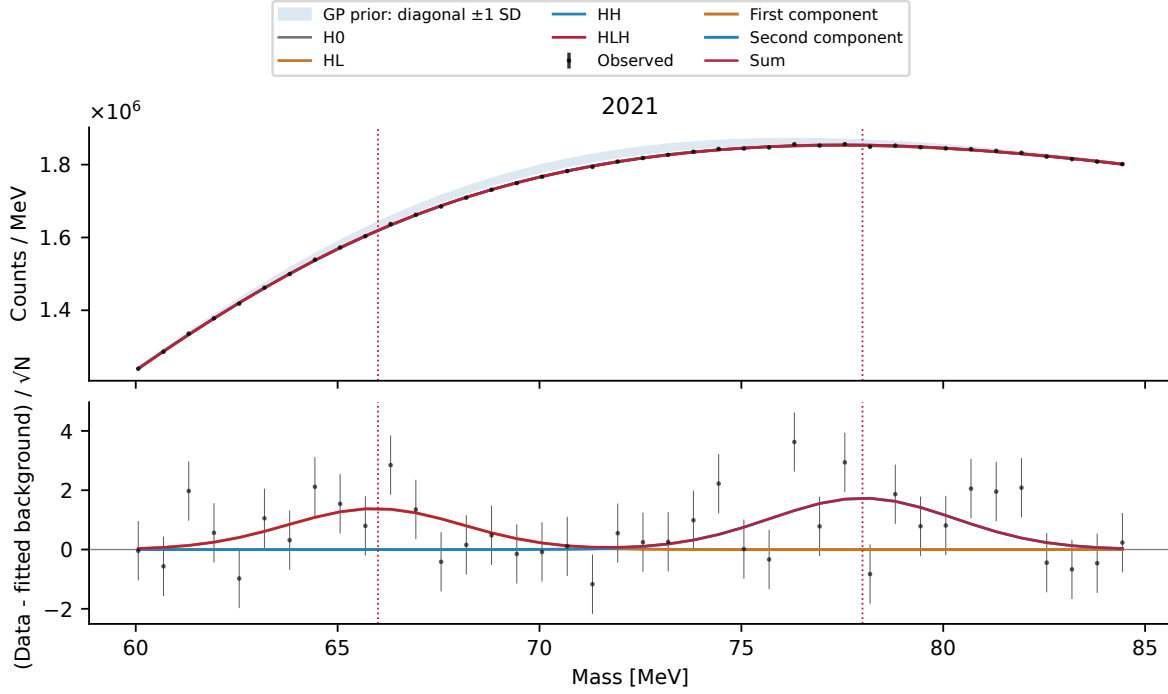
17. Mass surfaces with the region and background held fixed



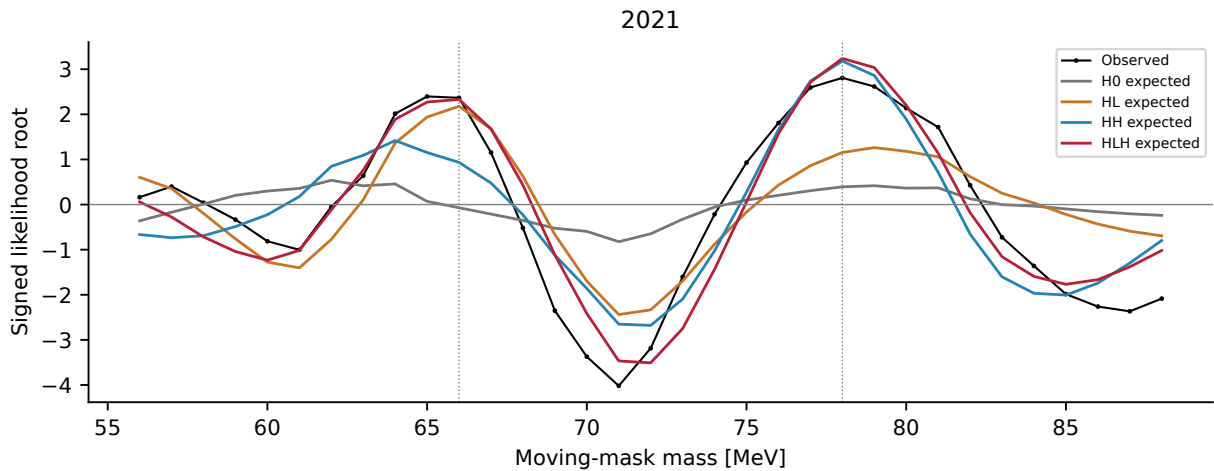
Each local rectangle is fitted using one fixed union region and one fixed midpoint kernel. Thus the plotted likelihood changes compare identical bins and background constraints as the two masses vary. The red cross marks the original catalogue pair, whose moving survey region is narrower. This changed fixed region is a stability diagnostic; its best cell is not silently substituted into the main catalogue.

18. The original 66/78 MeV proposal and the moving-mask dip

The original candidate neighborhood remains explicitly in the study. For all three campaigns, the fixed 66/78 MeV endpoint improvement is 1.92; for 2021 alone it is 2.49. The broader single-peak comparison is smaller where a nearby shifted single peak fits better. For the three-campaign 66/78 MeV fit, the full-region single comparison gives only $\Delta_{2/1} = 0.824$. The historical pair is therefore not the leading pair under this common-region policy.



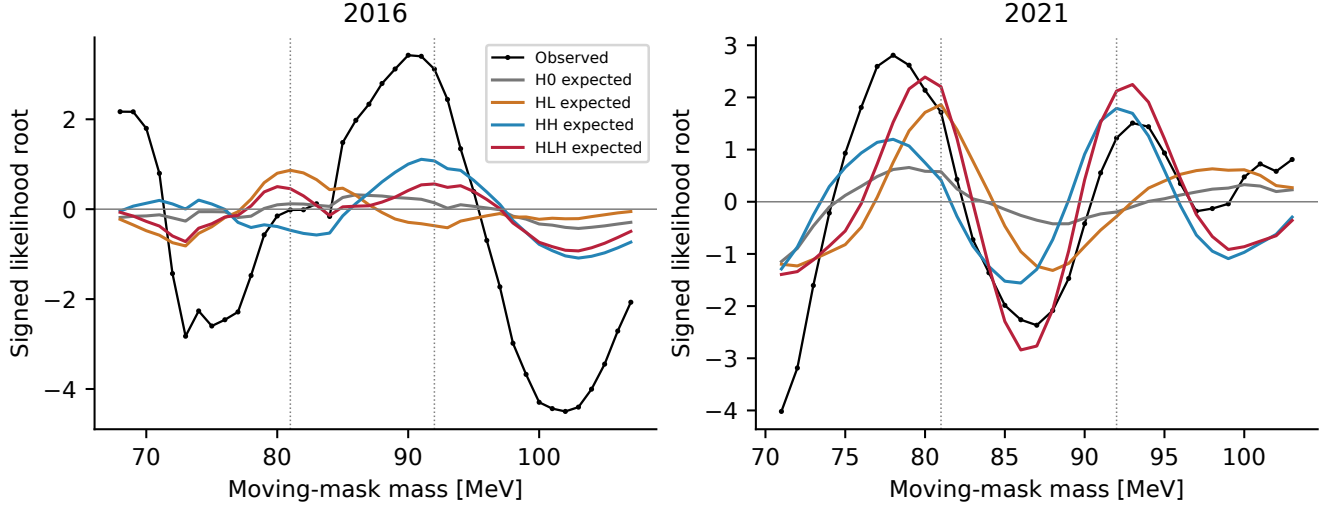
2021 counts at the historical pair. The dip bins remain in the same joint likelihood under all four hypotheses.



Deterministic moving-mask responses of the fitted full count spectra. Signal tails are included over the full support, and the GP is recomputed at every moving mask with its recorded kernel. The observed curve is a direct numerical replay. These posterior-fit comparisons reuse the data and are not independent likelihood terms.

19. Forward response at the maximal-available union leader

The union selects 81/92 MeV in 2016 + 2021. Each campaign below receives the same two fitted physical strengths. A good visual response in one campaign does not resolve incompatible relative rates in another.



The fitted H_0 , H_L , H_H , and H_{LH} expected count spectra are passed through the original moving-mask estimator. Their response includes sideband contamination from full-support signal tails. No fluctuations are drawn.

At 92 MeV, the shared two-component expectation predicts moving-mask roots about 0.56 in 2016 and 2.12 in 2021, compared with observed roots 3.12 and 1.22. The shared fit does not reproduce both campaign responses simultaneously.

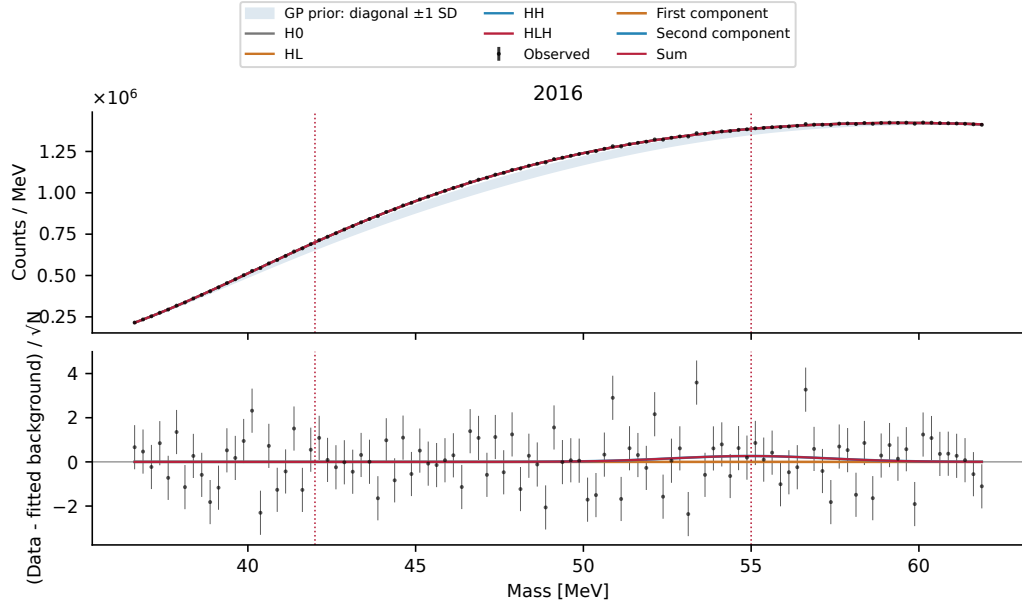
To construct a full fitted continuum, the common-region nuisance shift is extended through the GP cross-covariance. The inverse is applied only in the retained Poisson-whitened eigenbasis; this avoids an unstable solve when covariance modes have been compressed. The extension reproduces the fitted region, and positivity is checked over the full support. The resulting expected counts are a conditional reconstruction, not an independently established background truth.

The signal tails omitted from the external training assumption are retained in this forward diagnostic. At the nominal three-resolution padding, their fractions are small but nonzero and are listed in the campaign-yield ledger. Signal leakage can matter more for narrower padding. The forward scan consequently tests how a fitted count structure can produce neighboring positive and negative estimator responses; it does not count those responses as new observations.

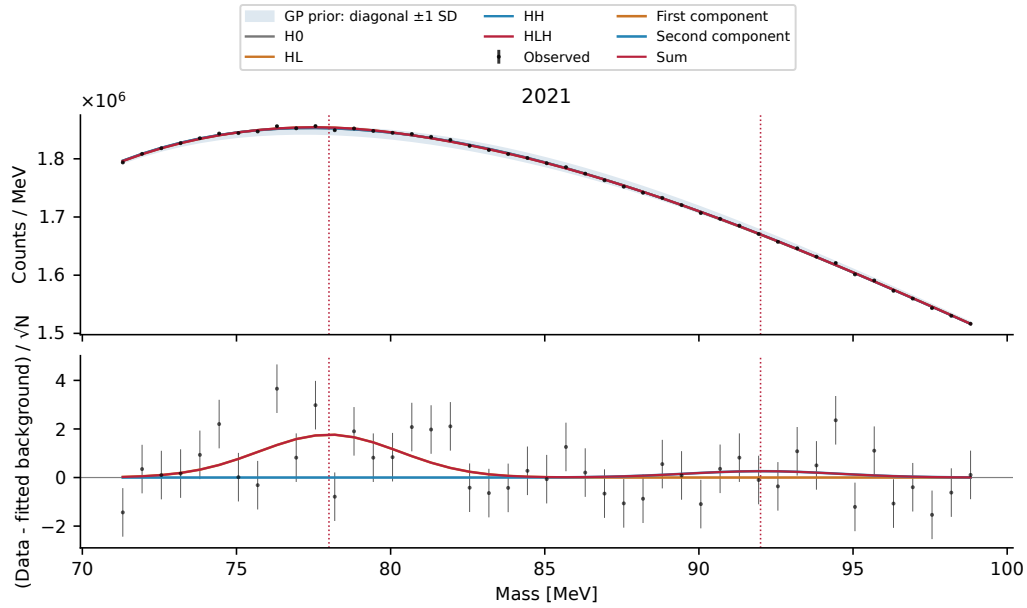
Campaign	N_1 shared	N_1 independent	N_2 shared	N_2 independent
2016	7022	28813	6415	33117
2021	13451	11114	16614	8027

20. Additional regions selected by the new GP local scores

The unusually small stress-centered probability near 42 MeV in 2016 is not accompanied by a strong two-positive-component count fit. The 92 MeV 2021 feature likewise adds little to the 78 MeV endpoint in this shared-background construction. Both diagnostics are shown rather than discarded because their pair improvement is small.



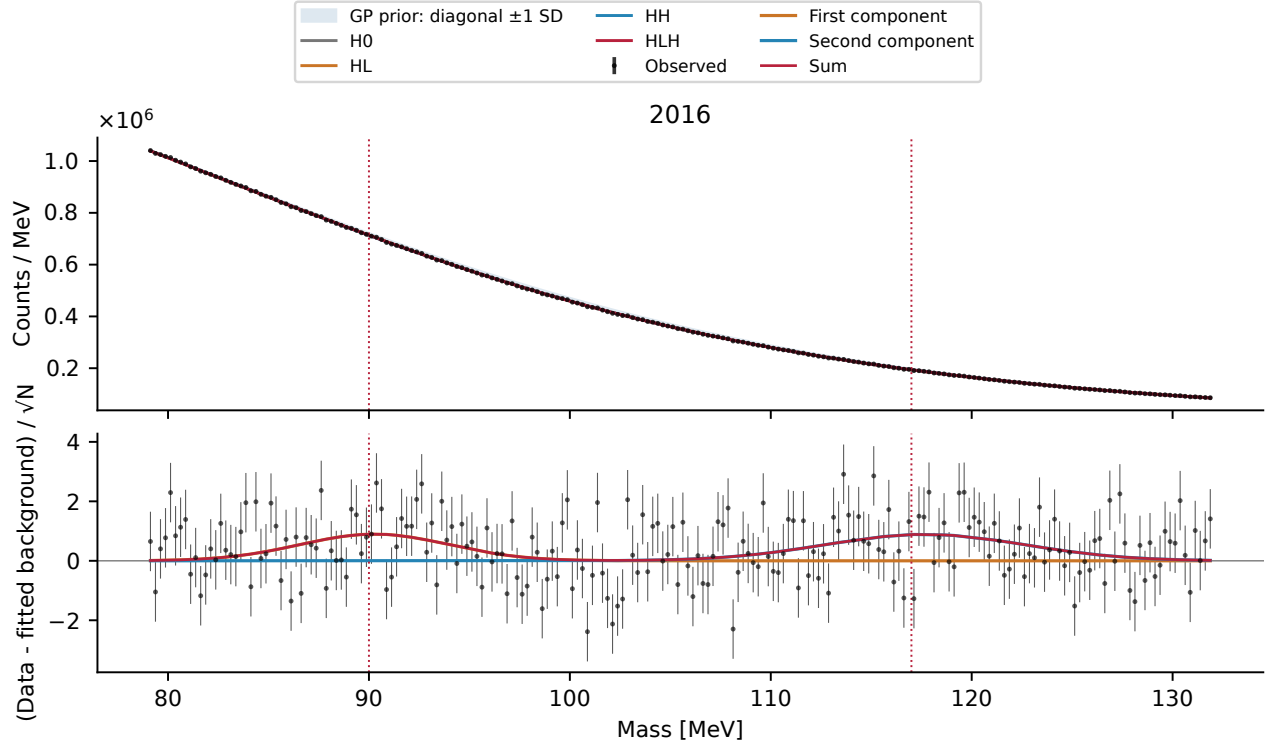
2016, 42/55 MeV: $Q_2 = 0.259$ and $q_{\text{add}} = 0$ at nominal padding. The large old conditional GP score is not a two-peak discovery statistic.



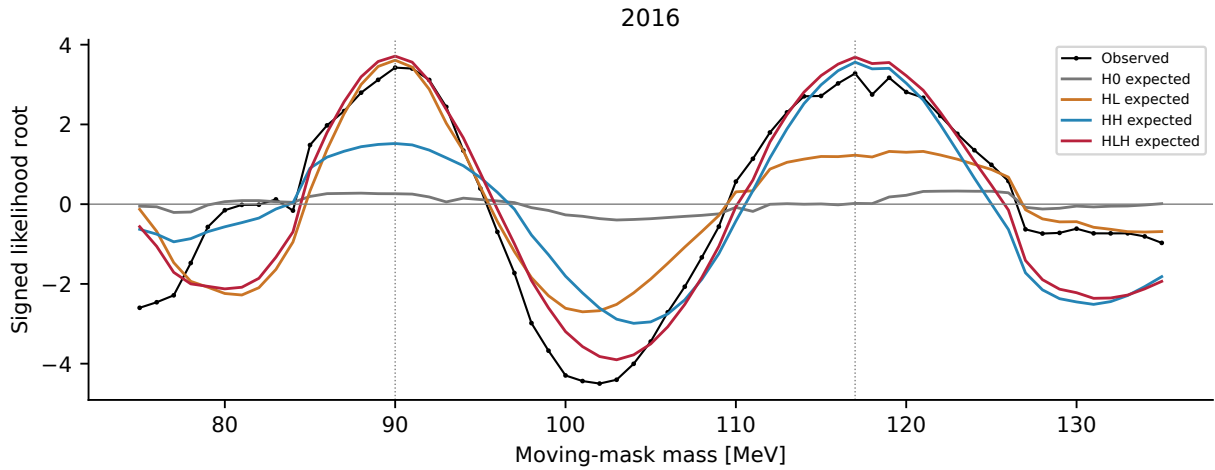
2021, 78/92 MeV: $Q_2 = 6.466$ and $q_{\text{add}} = 0.149$. The corresponding 2016+2021 endpoint improvement is 1.041. Detailed forward responses and joint profiles are included in the package.

21. A wider 2016 two-feature control

The v5.1.0 binning study identified competing raw maxima near 90 and 117 MeV. Their 27 MeV separation lies outside the initial pair survey, so this pair and neighboring targeted pairs are included explicitly. At nominal padding, the 90/117 MeV endpoint improvement is about 6.00 and $Q_2 = 20.18$. The broader best-single comparison is recorded separately in the complete catalogue.



The same original bins and background constraint are used under all four models across this wider interval. Its inclusion does not imply that every arbitrarily wide pair was searched.



Deterministic forward response at the wider pair. This is a morphology check under the chosen background model, not a calibrated comparison with the other scopes.

22. Reproducibility, validation, and remaining interpretation

What can be reproduced from this package

The standalone scripts use the pinned analysis-bin arrays, native density bins, resolution coefficients, kernel states, and conversion factors. Full ROOT inputs are identified by SHA-256 but are not needed to rerun the fits from these exported arrays. Original PDFs and their source identities are pinned as references. The full catalogue, signed two-versus-one comparisons, fitted components, positive expected full spectra, profile grids, numerical checks, and all plotted values are saved.

The build and reproduction instructions are in `README.md`. `protocol.json` records the finite scan and its exploratory additions. `inputs/manifest.json` identifies the sources, and `SHA256SUMS.txt` identifies the final study package. The final PDF is bound to numerical, semantic, portable-rebuild, and rendered-page QA records.

Numerical and scientific checks

Independent constrained optimization checks the four-hypothesis solver. Integer signal templates are checked against the original full-yield normalization. The original one-peak moving-mask roots are replayed, every refined coordinate is required in every eligible scope, and selected compressed covariance fits are checked against full rank. All accepted Poisson expectations and fitted backgrounds are positive. The fixed-region mass surfaces must retain exactly the same null likelihood under every mass pair. No failed fit is silently removed from the catalogue.

This is still a conditional observed-data analysis. Archived kernel hyperparameters retain their original narrower-mask data dependence; external-bin GP prediction here does not erase that provenance. The common-mask kernel prescription, background source family, covariance interpretation, campaign normalization, detector mass/width uncertainties, and event independence are not fully qualified here. No wider single-peak width is freely fitted; a broadened detector response is an additional possible explanation of close fitted components. The local reference p-values do not incorporate mass selection, scope selection, policy development, or a complete two-mass search maximum. Expected bands, power, coverage, and new calibrated local/global probabilities require subsequent work beyond this no-toy study.

References

- [1] *03: A concrete signal-assessment plan*, Section 5, HPS GPR Review and Signal Plan. Supplied local PDF and source, pinned in `inputs/`.
- [2] *Binning and significance fields*, independent report v5.1.0, 9 September 2026. Pinned nine-page report and source field arrays.
- [3] *HPS Gaussian-Process Resonance Search*, v5.0.2 review draft with the sigcorr audit, 9 September 2026. Pinned 186-page report.
- [4] S. Self and K. Liang, *Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions*, JASA 82 (1987) 605–610, Case 7. [Primary paper](#).
- [5] G. Cowan et al., *Asymptotic formulae for likelihood-based tests of new physics*, Eur. Phys. J. C 71 (2011) 1554. [arXiv:1007.1727](#).
- [6] E. Gross and O. Vitells, *Trial factors for the look elsewhere effect in high energy physics*, Eur. Phys. J. C 70 (2010) 525–530. [arXiv:1005.1891](#).
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